Exp No: 12	**Implementing security features and governance / administration practices in Zoho Creator.**

AIM:

To implement the security features of Zoho Creator (roles, permissions, record level and field level security, Deluge checks) and to follow the administration and governance practices such as the audit of the activities and the controlled sharing of the application.

ALGORITHM / PROCEDURE:

Security in Zoho Creator is applied in many layers. At the account level the admin can enable two factor authentication and restrict the login to trusted IP addresses. At the application level the access is controlled by **profiles and permissions** (who can view, add, edit or delete the data) and by **sharing settings** (who can open the application). At the data level the access can be limited by a criteria (record level) and by hiding the fields (field level). **Governance** means keeping control over the application by giving the minimum access needed, by keeping a record of the changes, and by reviewing the users regularly.

1. Create an application named *HR Records* with the form *Employee_Details* having the fields Name, Department, Phone and Salary.
2. Go to **Users & Control** → **Profiles/Roles** and create the profiles *HR_Manager* and *Employee*. Give the Employee profile only the permission to view and edit their own record, and no delete permission.
3. **Record level security:** in the permission of the Employee profile add the criteria *Added User = logged in user*, so that an employee can see only the record entered by himself.
4. **Field level security:** keep the field *Salary* visible only for the HR_Manager profile (from the field permission settings) and hide it for the others.
5. Add the users with the correct profile using **Share Application**. Do not give the admin profile to everyone. Use the minimum permission needed.
6. Add the Deluge scripts given below in the form workflows, to stop the user from editing the record of other persons and to hide the salary field in the form.
7. Open the account settings and turn on **two factor authentication**, and check the options to restrict the access by IP address (if the plan supports it).
8. Log in with the Employee user and the HR_Manager user one by one and check that each can see only the allowed records and fields.
9. Open the **Audit Trail / activity log** of the application and check that the changes made by the users with their date and time are recorded. Take the backup of the application data from the settings.
10. Note down the observations.

Script 1 - On Load of the form: hide the salary for the users other than admin

```
if(zoho.loginuser != zoho.adminuser)
{
    hide Salary;
}
```

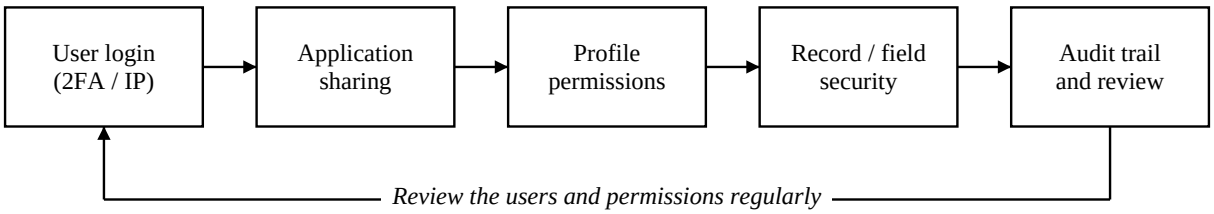
Script 2 - On Validate of Edit: allow the user to edit only his own record

```
if(zoho.loginuser != input.Added_User && zoho.loginuser != zoho.adminuser)
{
    alert "You are not allowed to edit this record.";
    cancel submit;
}
```

Note: the permissions of the profile are checked by Creator itself, so the scripts are only an additional layer and should not be the only protection. Hiding a field with a script does not remove its value, so the field level permission should also be set.

DIAGRAM:

Fig 1: Security and governance layers in a Zoho Creator application



RESULT:

Thus the security features of Zoho Creator were implemented using profiles, record level and field level security, and Deluge checks. Each user could access only the permitted records and fields, and the administration practices such as the audit trail, the data backup and the review of the users were followed.